

Uncertainty-Weighted Learnable Covariance Refinement for One-Bit Direction-of-Arrival Estimation With an Unknown Number of Sources

HYUK KWON¹, HANASOL JEON¹, YOUNGIN CHO¹, GYUNGMOK KIM¹, and HO YOUNG HWANG¹

¹Department of Computer Engineering, Kwangju University, Seoul 01897, Republic of Korea

Corresponding author: Ho Young Hwang (e-mail: hyhwang@kw.ac.kr).

Hyuk Kwon and Hanasol Jeon contributed equally to this work.

ABSTRACT One-bit analog-to-digital converters (ADCs) can reduce the power, cost, and data rate of sensor arrays, but they remove absolute amplitude information and alter spatial correlation through the arcsine law. Direction-of-arrival (DOA) estimation is further complicated when the number of sources is unknown and classical eigenvalue-based source enumeration is unreliable. We propose a learnable covariance probabilistic spectrum network (LCPSN) that jointly estimates the source count and DOAs from one-bit array snapshots. LCPSN recovers an arcsine-corrected correlation surrogate, refines it using a Transformer encoder and a Hermitian residual, and combines a Bartlett prior with spectrum mean and variance heads and an eigen-feature source-count head. For independent Gaussian snapshots, we derive a first-order, finite-snapshot variance approximation for arcsine-corrected correlation entries and use it to weight a covariance-consistency objective. On an eight-element uniform linear array with 200 snapshots and two to five sources, LCPSN obtains 97.4% source-count accuracy and a partially matched DOA mean absolute error (MAE) of 1.07° , compared with 60.1% and 3.86° for the best blind one-bit MUSIC pipeline and 94.1% and 2.46° for the strongest learned one-bit baseline. An ablation over three training seeds reduces MAE by 0.14° with uncertainty weighting, while the measured change in source-count accuracy is within seed variability. These results show improved performance under the specified blind evaluation protocol.

INDEX TERMS Array signal processing, deep learning, direction-of-arrival estimation, one-bit quantization, source enumeration, transformer, uncertainty quantification.

I. INTRODUCTION

DIRECTION-of-arrival (DOA) estimation with sensor arrays is a fundamental task in radar, sonar, wireless communication, and electronic surveillance. For large arrays, per-channel analog-to-digital converters (ADCs) can contribute substantially to power consumption, hardware cost, and data rate. ADC power generally increases with sampling rate and can rise steeply with resolution, depending on the converter architecture [1]. One-bit ADCs—essentially simple comparators—are therefore an attractive design point for massive and distributed apertures [2], [3], and DOA estimation from one-bit quantized array data has accordingly received sustained attention [4].

One-bit quantization, however, fundamentally changes the

statistical structure of the measurements. The sign operator removes absolute amplitude information; under the assumed Gaussian model, normalized spatial correlation remains identifiable, and the correlation of the quantized data relates to that of the analog data through the nonlinear arcsine (Van Vleck) law [5]. Subspace estimators such as MUSIC [7] and ESPRIT [8], or beamforming methods [9], can be applied to an arcsine-corrected covariance, but two practical obstacles remain. First, the corrected covariance is a plug-in estimate whose finite-snapshot error is strongly entry-dependent: the approximate variance depends on the sign-correlation level, so entries can differ in estimated reliability. Second, and often overlooked, the number of sources K is unknown in operation. Classical enumeration criteria such as

AIC and MDL [10] operate on covariance eigenvalues. In our implementation, MDL accuracy with the naive one-bit covariance is more than 30 percentage points below its full-precision counterpart; the arcsine correction alone does not restore MDL accuracy in this experiment. One-bit MUSIC formulations [11] assume K is given and thus sidestep this failure mode rather than solving it.

In parallel, learning-based DOA estimators have shown strong robustness to low SNR and model imperfections [12], [13], [14]. Most relevant to our setting, TransMUSIC [15] employs a Transformer-aided subspace framework for DOA estimation under low-resolution ADCs and also addresses source-number estimation. Deep unrolled sparse-recovery networks have also been demonstrated for one-bit DOA estimation [16]. In contrast to these works, our method explicitly derives and uses the entry-wise finite-snapshot uncertainty of the arcsine covariance estimate as a training weight, thereby incorporating the reliability of each recovered covariance entry into the learning objective.

We use a closed-form, first-order approximation to the finite-snapshot uncertainty of the one-bit correlation estimate as an explicit training weight. We propose the learnable covariance probabilistic spectrum network (LCPSN), which jointly performs source enumeration and DOA estimation from one-bit snapshots. Our contributions are as follows.

- 1) **Uncertainty-weighted covariance consistency.** We derive the entry-wise delta-method variance approximation U_0 for the arcsine-corrected correlation estimate from one-bit signs and use it to weight a covariance-consistency loss, so that the learnable refinement is anchored strongly to reliable entries and weakly to unreliable ones. We are not aware of prior one-bit DOA work that uses this variance as a weight in the training objective rather than as an analysis tool.
- 2) **Joint unknown- K and probabilistic spectrum estimation.** A Transformer encoder over covariance-row tokens refines the arcsine covariance through a learnable Hermitian residual; a Bartlett prior spectrum computed from the refined covariance is coupled residually into probabilistic spectrum heads that output both a mean spectrum and a per-angle variance, and an eigen-feature source-number head predicts K over $\{2, \dots, 5\}$.
- 3) **One-bit benchmark with blind enumeration.** We evaluate against three learned one-bit estimators and the full cross product of three covariance front-ends with three enumeration rules (MDL, AIC, oracle K), on matched test waveforms under one partial-matching protocol. The oracle- K rows diagnose source-count errors, but partial matching prevents them from serving as angular-error upper bounds. LCPSN reaches 97.4% enumeration accuracy and 1.07° MAE, exceeding even unquantized MUSIC under the same blind rule.
- 4) **Ablation of the uncertainty weighting.** Over three training seeds, the U_0 -weighted consistency improves angular accuracy at every SNR and every snapshot

count by more than the seed noise, and leaves source enumeration unchanged to within that noise. Both heads read features of the refined covariance, so the loss can affect either output; the observed source-count difference is within seed variability.

The remainder of this paper is organized as follows. Section II reviews related work. Section III formulates the one-bit signal model and derives the arcsine covariance uncertainty. Section IV details the proposed network and training objectives. Section V reports experiments and ablations, and Section VI concludes.

II. RELATED WORK

A. CLASSICAL DOA ESTIMATION AND SOURCE ENUMERATION

Subspace methods such as MUSIC [7] and ESPRIT [8] achieve super-resolution DOA estimation from the sample covariance [17], and their statistical performance is well characterized [18]. Sparsity-based [19] and gridless formulations provide high-resolution alternatives to classical subspace methods and have been widely studied for challenging array-processing regimes, including limited-snapshot and off-grid scenarios [20].

Subspace methods require the signal-subspace dimension, namely K ; other formulations may estimate K jointly or avoid an explicit subspace split. Information-theoretic criteria [10] and eigenvalue clustering heuristics can degrade when covariance estimates are distorted or based on few snapshots, as in the evaluated one-bit setting.

B. ONE-BIT ARRAY PROCESSING

The statistical foundation of one-bit processing is the Van Vleck arcsine law [5], complemented by Bussgang's decomposition [6], which relate the correlation of hard-limited Gaussian processes to that of the underlying analog processes. Early work established the feasibility of DOA estimation from one-bit measurements [4]. One line of methods performs sparse recovery directly on the one-bit data: binary iterative hard thresholding [21] and joint sparse representation [22] recover the angular spectrum from the sign measurements without forming a covariance. A second line operates on second-order statistics. One-bit MUSIC [11] exploits the fact that, under suitable assumptions, the signal and noise subspaces are preserved despite the scaling and identity-shift effects introduced by one-bit quantization, so that subspace methods can be applied directly to the one-bit covariance, given a known source number. In sparse-array settings, one-bit DOA methods instead use arcsine-law-based covariance recovery together with coarray covariance reconstruction to enable subspace processing on sparse geometries [23], and the estimation-theoretic limits of one-bit DOA have been characterized via the Cramér-Rao bound [24]. At the acquisition side, time-varying sampling thresholds, or dithering, have been shown to improve covariance recovery from one-bit sampled stationary signals by partially restoring amplitude-related information [25]; such techniques can be

combined with downstream DOA estimators, as also explored in learning-based one-bit DOA settings [16]. Many of these formulations assume a known source number or rely on conventional enumeration rules. Moreover, although arcsine-law covariance recovery is widely used—and is the starting point of our method—its estimated entry-dependent uncertainty is not incorporated into the training losses of the compared approaches.

C. LEARNING-BASED DOA ESTIMATION

Deep networks have been used to reconstruct or classify spatial spectra from covariance-related features, with demonstrated robustness to array imperfections [12] and low-SNR conditions [14]; DeepMUSIC, in particular, replaces the explicit spectrum computation with learned mappings [13].

Other DNN-based approaches estimate the number of sources and AoAs directly from array observations, showing performance advantages over conventional estimators in challenging settings [26]. Closest to our setting, TransMUSIC [15] uses a Transformer [27] to estimate the signal subspace from snapshots under low-resolution ADCs, performs gridless peak finding, and—like our method—also estimates the number of sources, through a dedicated classifier on the learned subspace. Deep unrolled sparse-recovery networks have been applied to one-bit DOA estimation, including settings with dithered one-bit measurements and covariance-recovery-based processing [16]; however, such methods do not use the closed-form arcsine covariance finite-snapshot variance as an entry-wise uncertainty weight in the training loss. Our work differs in two respects: we predict a probabilistic spectrum (mean and variance) rather than a point spectrum, in the spirit of heteroscedastic regression [28]; and—central to this paper—we inject the closed-form one-bit finite-snapshot uncertainty into the training objective itself, a reliability signal absent from the cited TransMUSIC and deep-unrolling formulations.

III. SIGNAL MODEL AND PROBLEM FORMULATION

A. ARRAY AND ONE-BIT MEASUREMENT MODEL

Consider a uniform linear array (ULA) of M omnidirectional sensors with half-wavelength spacing, receiving K narrow-band far-field signals from directions $\boldsymbol{\theta} = [\theta_1, \dots, \theta_K]^T$, $\theta_k \in (-90^\circ, 90^\circ)$. The analog snapshot at time t is

$$\mathbf{x}(t) = \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{n}(t), \quad t = 1, \dots, T, \quad (1)$$

where $\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_K)]$ with steering vector $[\mathbf{a}(\theta)]_m = e^{-j\pi(m-1)\sin\theta}$, $\mathbf{s}(t) \in \mathbb{C}^K$ collects mutually uncorrelated circular Gaussian source waveforms, and $\mathbf{n}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I})$ is spatially white noise. Each receiver chain quantizes the in-phase and quadrature components with a one-bit ADC at zero threshold (see Appendix A for the scaling convention):

$$\mathbf{y}(t) = \text{csgn}(\mathbf{x}(t)) := \text{sign}(\Re \mathbf{x}(t)) + j \text{sign}(\Im \mathbf{x}(t)). \quad (2)$$

Because $\text{sign}(\alpha u) = \text{sign}(u)$ for $\alpha > 0$, common positive scaling of the analog signal leaves the signs unchanged, so

absolute power cannot be recovered. The relative source-to-noise ratio can still influence normalized spatial correlation and thus the sign statistics under the assumed model.

The problem addressed in this paper is: given only the sign data $\{\mathbf{y}(t)\}_{t=1}^T$, jointly estimate the source number $K \in \{K_{\min}, \dots, K_{\max}\}$ and the DOAs $\{\theta_k\}_{k=1}^K$, without side information about K or the SNR.

B. ARCSINE COVARIANCE RECOVERY

Let $\tilde{\mathbf{s}}(t) = [\text{sign}(\Re \mathbf{x}(t))^T, \text{sign}(\Im \mathbf{x}(t))^T]^T \in \{\pm 1\}^{2M}$ denote the real-augmented sign vector, and define the empirical sign correlation

$$\hat{\boldsymbol{\tau}} = \frac{1}{T} \sum_{t=1}^T \tilde{\mathbf{s}}(t) \tilde{\mathbf{s}}(t)^T. \quad (3)$$

For zero-mean jointly Gaussian inputs, the arcsine law [5] gives $\mathbb{E}[\tilde{s}_i \tilde{s}_j] = \frac{2}{\pi} \arcsin(c_{ij})$, where c_{ij} is the normalized correlation of the corresponding analog components. Inverting yields the arcsine (Van Vleck) correction

$$\hat{\mathbf{C}} = \sin\left(\frac{\pi}{2} \hat{\boldsymbol{\tau}}\right), \quad (4)$$

applied entry-wise. In the complex domain, the equivalent correction used by our network normalizes the quantized sample covariance $\mathbf{R}_z = \frac{1}{T} \sum_t \mathbf{y}(t) \mathbf{y}(t)^H$ to unit diagonal, $\mathbf{C}_z = \mathbf{D}^{-1/2} \mathbf{R}_z \mathbf{D}^{-1/2}$ with $\mathbf{D} = \text{diag}(\Re \mathbf{R}_z)$, and forms

$$\mathbf{R}_0 = \sin\left(\frac{\pi}{2} \Re \mathbf{C}_z\right) + j \sin\left(\frac{\pi}{2} \Im \mathbf{C}_z\right), \quad (5)$$

followed by Hermitian symmetrization $\mathbf{R}_0 \leftarrow \frac{1}{2}(\mathbf{R}_0 + \mathbf{R}_0^H)$. Equation (5) applies the correction (4) entry-wise to the real and imaginary parts of the normalized complex correlation, and agrees with the real-augmented construction at the population level under the circular Gaussian assumptions. At finite T , applying the sine to an average of real/imaginary correlations need not equal averaging their separately corrected values. Note that for sign data $\text{diag}(\mathbf{R}_z) = 2$ holds deterministically, so the normalization amounts to a constant scaling; its role is to make $\mathbf{R}_0$ invariant to the quantizer amplitude convention rather than to adapt to the data. $\mathbf{R}_0$ estimates normalized spatial correlation under the Gaussian assumptions, but finite- T plug-in error remains; the analog covariance scale is not recovered.

C. FINITE-SNAPSHOT UNCERTAINTY OF THE ARCSINE ESTIMATE

We estimate entry-wise uncertainty of $\hat{\mathbf{C}}$. Assuming independent, identically distributed snapshots, each $\hat{\tau}_{ij}$ is an average of T sign products and $\text{var}(\hat{\tau}_{ij}) = (1 - \tau_{ij}^2)/T$. Applying the delta method and replacing the unknown τ_{ij} with its sample estimate gives the following first-order plug-in approximation:

$$[\mathbf{U}_0]_{ij} = \left(\frac{\pi}{2}\right)^2 \cos^2\left(\frac{\pi}{2} \hat{\tau}_{ij}\right) \frac{1 - \hat{\tau}_{ij}^2}{T}, \quad (6)$$

where $\hat{\tau}$ is first clipped to $|\hat{\tau}_{ij}| \leq 1 - 10^{-6}$ and the result is clamped to $[\varepsilon, u_{\max}]$ with $\varepsilon = 10^{-6}$, $u_{\max} = 10$. On

the diagonal, $\hat{\tau}_{ii} = 1$ makes the delta-method variance vanish identically, so $[\mathbf{U}_0]_{ii} = \varepsilon$ by construction; through the consistency objective in Section IV-E this acts as a strong unit-diagonal anchor on the refined covariance, which is consistent with the sign data carrying no amplitude information on the diagonal. Within the first-order approximation (6), the estimated variance tends to zero as $|\hat{\tau}_{ij}|$ approaches one and decreases as $1/T$ for independent snapshots. Dependence between snapshots would alter this $1/T$ scaling. $\mathbf{U}_0 \in \mathbb{R}^{2M \times 2M}$ is computed on the real-augmented blocks, so it provides separate reliabilities for the real and imaginary parts of every covariance entry. We emphasize that $\mathbf{U}_0$ is not an input feature to the network; it enters only the training objective in Section IV-E.

IV. PROPOSED METHOD

Fig. 1 summarizes the six stages of LCPSN.

LCPSN is organized into six stages, and we use the same names for them in the text, in the figures, and in the loss definitions of Section IV-E:

- (S1) arcsine covariance recovery, the closed-form front end of Sections III-B and III-C, which produces $\mathbf{R}_0$ and $\mathbf{U}_0$ and holds no learned parameter;
- (S2) covariance tokenization and Transformer encoding, which maps the M rows of $\mathbf{R}_0$ to tokens and returns contextualized row features together with the pooled encoder feature $\bar{\mathbf{h}}$;
- (S3) Hermitian residual refinement, which adds a learned Hermitian residual to $\mathbf{R}_0$ and outputs the refined covariance $\mathbf{R}_{\text{ref}}$;
- (S4) the probabilistic spectrum heads, which couple the Bartlett prior spectrum S_b of $\mathbf{R}_{\text{ref}}$ with $\bar{\mathbf{h}}$ and emit the spectrum mean $\boldsymbol{\mu}$ and the per-angle variance $\boldsymbol{\sigma}^2$;
- (S5) the eigen-feature source-number head, abbreviated K -head, which reads the same fused feature plus the eigen-features of $\mathbf{R}_{\text{ref}}$ and emits the source-number logits;
- (S6) the peak decoder, which turns $(\boldsymbol{\mu}, \hat{K})$ into the reported DOAs.

Stages S1–S3 act on covariances, S4–S5 are the prediction heads, and S6 is parameter-free. Throughout, $d = 96$ denotes the model width and $G = 360$ the size of the angular grid

$$\theta_g = -90^\circ + g \cdot \frac{180^\circ}{G-1}, \quad g = 0, \dots, G-1,$$

a uniform 0.5014° spacing over the closed interval $[-90^\circ, 90^\circ]$. This grid defines both the supervision target of Section IV-E and the angles reported by the peak decoder, so that bin index g denotes the same direction at training and at test time. The Bartlett prior uses a half-open grid while the targets and decoder use the closed grid; this implementation mismatch should be removed or quantified in a controlled test (Appendix B).

Figs. 2–4 detail the pipeline in the order a measurement traverses it. Fig. 2 recapitulates the arcsine covariance recovery of Sections III-B and III-C, i.e., stage S1; Fig. 3 covers stages S2 and S3, which share the token features; and Fig. 4

covers stages S4 and S5 together with the peak decoder S6 that turns the spectrum into angles.

A. COVARIANCE TOKENIZATION AND TRANSFORMER ENCODING (S2)

The row-wise covariance tokenizer embeds the m -th row of $\mathbf{R}_0$ as a token

$$\mathbf{t}_m = \mathbf{W} \text{LN}([\Re(\mathbf{R}_0)_{m,:}, \Im(\mathbf{R}_0)_{m,:}]) + \mathbf{b} \in \mathbb{R}^d, \quad (7)$$

where LN is layer normalization with learnable affine parameters, $\mathbf{W} \in \mathbb{R}^{d \times 2M}$, and $\mathbf{b} \in \mathbb{R}^d$. The M tokens are processed by an L -layer pre-norm Transformer encoder [27] ($L = 4, 8$ heads, feed-forward width 256, GELU activations, dropout 0.1), in which each sublayer applies layer normalization before the multi-head self-attention or feed-forward block and adds the result residually. This yields contextualized row features $\mathbf{h}_1, \dots, \mathbf{h}_M$ and a pooled encoder feature $\bar{\mathbf{h}} = \text{GELU}(\mathbf{W}_p \text{LN}(\frac{1}{M} \sum_m \mathbf{h}_m) + \mathbf{b}_p) \in \mathbb{R}^d$, followed by dropout during training. The pooling is therefore a mean over the M row features followed by a normalized affine–GELU projection, not a bare average.

B. HERMITIAN RESIDUAL REFINEMENT (S3)

Each token feature is decoded to a complex row correction $\boldsymbol{\delta}_m \in \mathbb{C}^M$ by a two-layer MLP, forming the residual $\boldsymbol{\Delta} \in \mathbb{C}^{M \times M}$ row-wise. The refinement is constrained to the Hermitian manifold by symmetrization and applied through a learnable gate α (initialized to 0.1):

$$\mathbf{R}_{\text{ref}} = \mathbf{R}_0 + \alpha \cdot \frac{1}{2}(\boldsymbol{\Delta} + \boldsymbol{\Delta}^H), \quad (8)$$

followed by a final Hermitian symmetrization. The residual form keeps the physically grounded $\mathbf{R}_0$ as the operating point and lets the network learn only a correction, which we found trained more stably than predicting $\mathbf{R}_{\text{ref}}$ outright. Fig. 3 summarizes stages S2 and S3 together, since the refinement consumes exactly the token features produced by the encoder.

C. PROBABILISTIC SPECTRUM HEADS (S4)

From the refined covariance we compute the normalized Bartlett prior spectrum [30] S_b on the grid $\{\theta_g\}_{g=0}^{G-1}$,

$$S_b(\theta_g) = \frac{q_g}{\max_{g'} q_{g'}}, \quad (9)$$

$$q_g = \max(\mathbf{a}^H(\theta_g) \mathbf{R}_{\text{ref}} \mathbf{a}(\theta_g), \epsilon_b),$$

which serves as a physics-based prior: it is low-resolution, but requires no assumed source number; its array manifold uses the stated ULA model. The floor $\epsilon_b = 10^{-8}$ does more than guard against round-off. The refinement (8) constrains $\mathbf{R}_{\text{ref}}$ to be Hermitian but not positive semidefinite, so the quadratic form can be negative; the floor keeps S_b in $(0, 1]$ and thus a valid input to the spectrum heads.

The spectrum heads consume the fused feature $\mathbf{z} = [\bar{\mathbf{h}}; \beta S_b] \in \mathbb{R}^{d+G}$, where β is a learnable coupling (initialized to 0.75), and produce a probabilistic spectrum. The same scalar β is shared by this concatenation and the residual

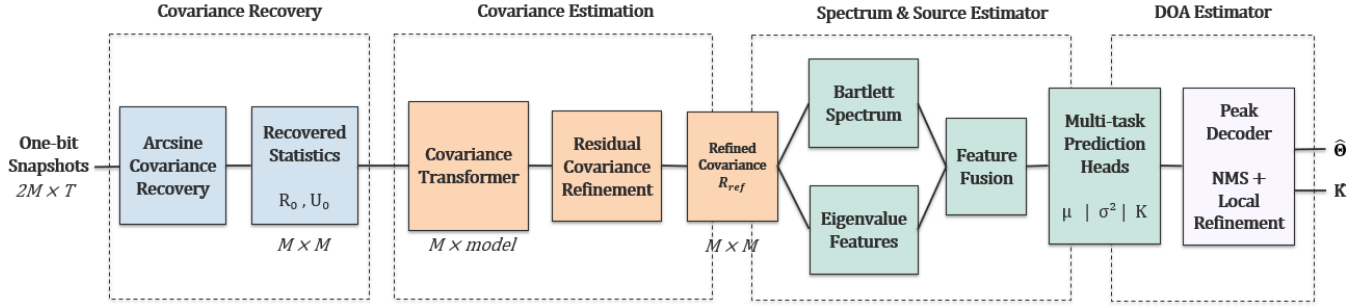


FIGURE 1. Overall pipeline of the proposed LCPSN, drawn as the six stages S1–S6 named in the text. *Arcsine covariance recovery* (S1) maps the one-bit signs to the arcsine covariance $\mathbf{R}_0 \in \mathbb{C}^{M \times M}$ and its closed-form uncertainty $\mathbf{U}_0 \in \mathbb{R}^{2M \times 2M}$, the latter being consumed by the training objective (19) only and never by the network. *Covariance tokenization and Transformer encoding* (S2) turns the M rows of $\mathbf{R}_0$ into tokens of width d and returns the pooled encoder feature $\bar{\mathbf{h}}$; *Hermitian residual refinement* (S3) adds the symmetrized residual Δ through the learnable gate α to give the refined covariance $\mathbf{R}_{\text{ref}}$. The *probabilistic spectrum heads* (S4) read $\mathbf{z} = [\bar{\mathbf{h}}; \beta S_b]$, where S_b is the Bartlett prior spectrum of $\mathbf{R}_{\text{ref}}$, and emit the spectrum mean μ and the per-angle variance σ^2 ; the *eigen-feature source-number head* (S5) reads $\mathbf{z}$ together with the eigen-features of $\mathbf{R}_{\text{ref}}$ and emits the source-number logits. The *peak decoder* (S6) returns $\hat{K}$ and the DOAs $\hat{\theta}_1, \dots, \hat{\theta}_{\hat{K}}$.

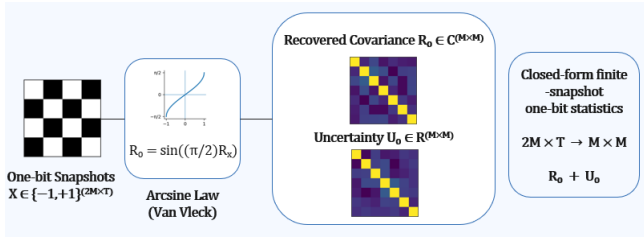


FIGURE 2. Stage S1, arcsine covariance recovery. The one-bit snapshots $\mathbf{y}(1), \dots, \mathbf{y}(T)$, equivalently the real-augmented signs $\tilde{\mathbf{s}}(t) \in \{\pm 1\}^{2M}$, give the sign-correlation matrix $\hat{\mathbf{r}} \in \mathbb{R}^{2M \times 2M}$; the arcsine law (4) inverts the quantizer to $\mathbf{R}_0 \in \mathbb{C}^{M \times M}$, and the delta method (6) supplies the entry-wise uncertainty $\mathbf{U}_0 \in \mathbb{R}^{2M \times 2M}$ at no extra cost. Both are closed form; neither involves a learned parameter. $\mathbf{U}_0$ is used only by the training objective (19) and is never an input to the network.

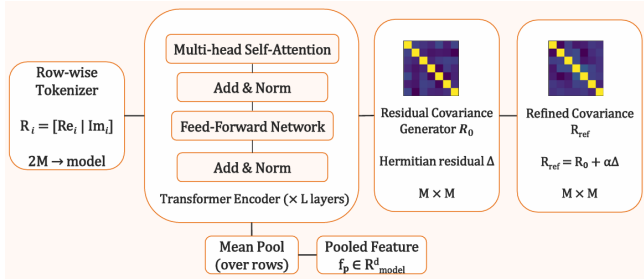


FIGURE 3. Stages S2 and S3: covariance tokenization, Transformer encoding, and Hermitian residual refinement. Each row of $\mathbf{R}_0$ becomes one token of width d in (7); the L -layer pre-norm encoder contextualizes the M tokens and also yields the pooled encoder feature $\bar{\mathbf{h}}$ used by the heads of stages S4 and S5; the decoded row corrections form Δ , which is Hermitian-symmetrized and added back through the learnable gate α in (8) to give $\mathbf{R}_{\text{ref}}$.

coupling in (10). The fused feature is the only input of the two spectrum heads: the eigen-features of Section IV-D are read by the K -head alone. This follows the general idea of heteroscedastic uncertainty modeling, in which the network predicts both a mean response and an input-dependent variance, as commonly used in probabilistic deep regression [28]; here we adapt it to angular spectrum prediction rather than

treating [28] as a DOA-specific uncertainty method:

$$\tilde{S}_b = \text{clip}(S_b, \epsilon_s, 1 - \epsilon_s), \quad (10)$$

$$\mu = \text{sigm}(f_\mu(\mathbf{z}) + \beta \logit(\tilde{S}_b)) \in (0, 1)^G,$$

$$\sigma^2 = \text{softplus}(\mathbf{v}) + \epsilon_v, \quad \mathbf{v} = \text{clip}(f_v(\mathbf{z}), -10, 5), \quad \epsilon_v = 0.05 \quad (11)$$

where $\text{sigm}(\cdot)$ is the logistic sigmoid, f_μ, f_v are two-hidden-layer MLPs (width 512), and $\epsilon_s = 10^{-4}$. The clip on S_b is not optional, and it binds at both ends: the normalization in (9) makes $\max_g S_b$ exactly one, so $\logit(S_b)$ diverges at the peak bin of every scene, while the floor ϵ_b leaves S_b free to fall to $O(10^{-10})$ in deep nulls, where the logit diverges the other way. The clip on $\mathbf{v}$ in (11) is, by contrast, an inactive numerical safeguard at the reported operating point. The residual coupling $\beta \logit(\tilde{S}_b)$ in (10) anchors the learned spectrum to the Bartlett prior spectrum. The variance head provides a learned per-angle dispersion parameter used by the Gaussian negative log-likelihood (16a); its calibration as a predictive confidence measure is not assessed here.

D. EIGEN-FEATURE SOURCE-NUMBER HEAD (S5)

Source enumeration uses both the learned features and explicit spectral statistics of the refined covariance: with eigenvalues $\lambda_1 \geq \dots \geq \lambda_M$ of $\mathbf{R}_{\text{ref}}$, which need not be positive semidefinite, eigenvalues are floored to a positive numerical threshold of 10^{-6} before taking logs; the eigen-features are the resulting log-eigenvalues $\log \lambda$ and the $M - 1$ consecutive log-eigenvalue gaps $\log \lambda_{1:M-1} - \log \lambda_{2:M}$, so that the input of the K -head is $[\mathbf{z}; \log \lambda; \log \lambda_{1:M-1} - \log \lambda_{2:M}] \in \mathbb{R}^{d+G+2M-1}$, and a two-layer MLP outputs the logits ℓ_K over $K \in \{K_{\min}, \dots, K_{\max}\} = \{2, \dots, 5\}$. Unlike fixed AIC/MDL-type rules, this head learns the mapping from the quantization-distorted eigen-spectrum, refined by the network, to the source count.

E. TRAINING OBJECTIVES

The supervision target is a soft spectrum with Gaussian lobes at the true DOAs, $T_g = \max_k \exp(-(\theta_g - \theta_k)^2 / 2\sigma_a^2)$,

whose width σ_a is annealed from 2.0° to 1.2° over training (constant during the first 15% of epochs, then linear), implementing a coarse-to-fine curriculum.

1) Spectrum losses

The angular grid is sparse: a scene has at most five source angles among $G = 360$ bins, and each initial 2.0° Gaussian lobe has fewer than ten bins above $T_g = \frac{1}{2}$. This motivates weighting positive responses rather than treating all grid bins equally. The main term is therefore a focal binary cross-entropy [31] with positive-class weighting,

$$p_g = T_g \mu_g + (1 - T_g)(1 - \mu_g), \quad (12)$$

$$\mathcal{L}_{\text{focal}} = \frac{1}{G} \sum_{g=0}^{G-1} (1 - p_g)^\gamma [-w_{\text{pos}} T_g \log \mu_g - (1 - T_g) \log(1 - \mu_g)], \quad (13)$$

where p_g is the soft-target generalization of the focal “probability of the correct class”, $\gamma = 2$ down-weights bins the network already gets right, and $w_{\text{pos}} = 10$ multiplies the positive term only, following the `pos_weight` convention, so as to compensate the positive-class scarcity without rescaling the negative term. The two factors address different failures: without w_{pos} the network under-responds at true DOAs, and without γ the easy empty bins dominate the average once the lobes are roughly placed.

Peak counting, however, is not sensitive to the average loss but to the largest spurious response, which the focal term penalizes only in proportion to its own probability. We therefore add an explicit false-peak penalty on the empty region,

$$\mathcal{L}_{\text{fp}} = \frac{1}{G} \sum_{g: T_g < 0.05} \mu_g^2, \quad (14)$$

which penalizes high false responses more heavily than small ones for the same total response mass. The spectrum loss is

$$\mathcal{L}_{\text{spec}} = \mathcal{L}_{\text{focal}} + 0.35 \mathcal{L}_{\text{fp}}, \quad (15)$$

where the 0.35 is applied directly rather than through a λ because the two terms supervise the same output. The probabilistic pair (μ, σ^2) is additionally trained by a Gaussian negative log-likelihood

$$\mathcal{L}_{\text{gnll}} = \frac{1}{2G} \sum_{g=0}^{G-1} \left[\frac{(T_g - \mu_g)^2}{\sigma_g^2} + \log \sigma_g^2 \right], \quad (16a)$$

$$\mathcal{L}_{\text{var}} = \frac{1}{G} \sum_{g=0}^{G-1} v_g^2, \quad (16b)$$

where $\mathbf{v}$ is the (clipped) pre-activation of the variance head defined in (11). Penalizing the pre-activation rather than $\log \sigma_g^2$ pulls the predicted variance toward its rest point $\text{softplus}(0) + \varepsilon_v \approx 0.74$, preventing the variance head from collapsing or diverging on ambiguous quantized inputs.

2) Source-number loss

Enumeration is supervised as a $|\mathcal{K}|$ -way classification over $\mathcal{K} = \{K_{\min}, \dots, K_{\max}\}$ with label smoothing,

$$\mathcal{L}_K = - \sum_{k \in \mathcal{K}} q_k \log [\text{softmax}(\ell_K)]_k, \quad (17)$$

$$q_k = (1 - \epsilon) \mathbf{1}[k = K] + \frac{\epsilon}{|\mathcal{K}|}, \quad (18)$$

with $\epsilon = 0.1$ and ℓ_K the K -head logits. The confusion matrices in Section V show that most errors involve adjacent source counts, especially dense scenes. Label smoothing is used to discourage overconfident predictions, although its separate effect is not isolated by an ablation.

3) Uncertainty-weighted covariance consistency

The refinement (8) must not drift arbitrarily from the physically recovered $\mathbf{R}_0$ —but the appropriate stiffness of this anchor differs per entry, exactly as quantified by $\mathbf{U}_0$ in (6). Splitting $\mathbf{U}_0$ into its $M \times M$ blocks $\mathbf{U}_{rr}, \mathbf{U}_{ri}, \mathbf{U}_{ir}, \mathbf{U}_{ii}$ (real/imaginary sign streams), working weights for the real and imaginary parts of each complex entry are formed as $\mathbf{V}_{\Re} = \frac{1}{4}(\mathbf{U}_{rr} + \mathbf{U}_{ii})$ and $\mathbf{V}_{\Im} = \frac{1}{4}(\mathbf{U}_{ri} + \mathbf{U}_{ir})$; the factor $\frac{1}{4}$ is derived in Appendix C. We define

$$\mathcal{L}_{\text{ucov}} = \frac{1}{M^2} \sum_{i,j} \left[\frac{(\Re \Delta R_{ij})^2}{[\mathbf{V}_{\Re}]_{ij}} + \frac{(\Im \Delta R_{ij})^2}{[\mathbf{V}_{\Im}]_{ij}} \right], \quad (19)$$

where $\Delta \mathbf{R} = \mathbf{R}_{\text{ref}} - \text{sg}(\mathbf{R}_0)$, $\text{sg}(\cdot)$ stops gradients into the plug-in estimate, and the denominators are floored at $\epsilon_u = 10^{-6}$. This floor is not optional: on the diagonal $\hat{\tau}_{ij} = 1$ for $i = j$, so $\cos^2(\frac{\pi}{2} \hat{\tau}_{ij}) = 0$ and the delta-method variance in (6) vanishes exactly, which would make (19) divide by zero. Equation (19) is a diagonal, variance-weighted anchor: entries whose arcsine recovery is statistically reliable (small $\mathbf{U}_0$) hold the refinement tightly, while unreliable entries grant the network freedom to deviate. This is the mechanism by which the closed-form one-bit uncertainty enters learning.

4) Total objective

$$\mathcal{L} = \mathcal{L}_{\text{spec}} + \lambda_{\text{gnll}} \mathcal{L}_{\text{gnll}} + \lambda_K \mathcal{L}_K + \lambda_v \mathcal{L}_{\text{var}} + \lambda_u \mathcal{L}_{\text{ucov}}, \quad (20)$$

with $\lambda_{\text{gnll}} = 10^{-2}$, $\lambda_K = 0.5$, $\lambda_v = 5 \times 10^{-3}$, and $\lambda_u = 10^{-3}$ unless stated otherwise.

F. INFERENCE AND PEAK DECODER (S6)

As illustrated in Fig. 4, at test time $\hat{K} = \arg \max$ of the K -head plus $K_{\min}$, and the peak decoder returns as DOAs the top- $\hat{K}$ local maxima of μ subject to a minimum separation of 2° (with a greedy fallback if fewer peaks exist); the decoder holds no parameter and performs no sub-grid interpolation, so a reported angle is always a grid point θ_g . That floor is set by the spectral resolution implied by the training lobe width, not by the 8° separation of the test scenes; using the latter would hand the decoder a prior on the test condition that the regressor baselines do not receive. The forward pass includes an eigendecomposition for the K -head, but

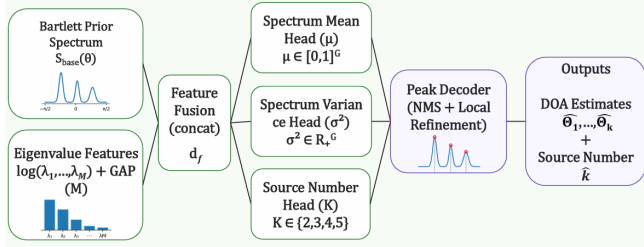


FIGURE 4. Stages S4–S6, the inference path. The Bartlett prior spectrum S_b computed from $\mathbf{R}_{\text{ref}}$ is concatenated with the pooled encoder feature $\bar{\mathbf{h}}$ through the learnable coupling β to form $\mathbf{z}$; the two spectrum heads emit the spectrum mean μ and the per-angle variance σ^2 from $\mathbf{z}$ alone, while the K -head additionally reads the eigen-features of $\mathbf{R}_{\text{ref}}$ and emits the source-number logits. The peak decoder takes the top- $\hat{K}$ local maxima of μ with at least 2° separation and returns $\hat{K}$ and $\hat{\theta}_1, \dots, \hat{\theta}_{\hat{K}}$.

requires no iterative peak optimization or grid refinement; a single forward pass produces $\hat{K}$, the DOAs, and per-angle confidence σ^2 .

V. EXPERIMENTS

A. SETUP

1) Data

We use a ULA with $M = 8$ half-wavelength-spaced sensors. Sources are mutually uncorrelated circular Gaussian waveforms; the source number is drawn from $K \in \{2, 3, 4, 5\}$ with a minimum angular separation of 8° , and DOAs are sampled uniformly over $(-90^\circ, 90^\circ)$. Training and validation sets contain 256,000 and 25,600 scenes at $T = 200$ with SNR mixed over $\{0, 5, 10, 15, 20\}$ dB. Test scenes are generated on the fly from seeds disjoint from training, 25,600 per operating point, and every method sees the same waveforms at a given point rather than merely the same random seed. We sweep SNR from -10 to 20 dB at $T = 200$, and snapshots from $T = 50$ to $T = 1000$ at 10 dB.

2) Training

LCPSN is trained for 100 epochs with AdamW [29] (learning rate 2×10^{-4} , weight decay 2×10^{-4}), cosine-annealed schedule, batch size 128, and gradient clipping at norm 1.0. Model selection uses a validation monitor combining the spectrum, K , and consistency terms. Each learned method is trained with seeds 0, 1, and 2 using its model-specific optimization recipe, and all reported learned results are means over these three training seeds. The training datasets are generated independently but use the same array model, SNR set, source-count distribution, minimum angular separation, sample counts, and fixed data-generation seeds. Across-seed variability for LCPSN is reported in the text and used as a descriptive reference when interpreting the ablation.

3) Metrics

All methods are scored under $\hat{K}$ -decoding: a method reports the $\hat{K}$ angles it actually decoded, and these are matched to the K true angles by the permutation minimizing the angular error over $n = \min(\hat{K}, K)$ pairs, with mod- π wrapping.

We report (i) the source-number accuracy $\Pr[\hat{K} = K]$; and (ii) an RMSPE evaluated under minimum-error partial matching. Following the periodic and permutation-invariant treatment used in prior learning-based DOA work such as SubspaceNet [32], angular differences are wrapped modulo π , and the assignment is permutation invariant. For unequal cardinalities, we select the minimum-error matching of $n = \min(\hat{K}, K)$ pairs:

$$\text{RMSPE} = \min_{\mathcal{M}_n} \left[\frac{1}{n} \sum_{(i,j) \in \mathcal{M}_n} \text{wrap}_\pi(\hat{\theta}_i - \theta_j)^2 \right]^{1/2},$$

where $\mathcal{M}_n$ denotes a one-to-one matching of n predicted and true angles, and wrap_π maps an angular difference into $[-\pi/2, \pi/2)$. The matched MAE is evaluated using the same assignment that minimizes RMSPE. We additionally report (iii) the success rate $\text{SR}@5^\circ$, defined as the fraction of all evaluated scenes whose matched RMSPE is below 5° . Conditional metrics restricted to scenes with $\hat{K} = K$ are computed as diagnostics but are not included in the main comparison. Scene-level values are averaged over scenes, so each scene carries equal weight regardless of its source count; the reported RMSPE is thus a mean of per-scene root-mean-square errors, not a pooled quadratic mean. Because only $n = \min(\hat{K}, K)$ angles are scored, a method that undercounts is not charged for the sources it never reported; this is a property of $\hat{K}$ -decoding that we return to when reading Table 1.

4) Baselines

We compare against three learned estimators and nine classical ones. The learned baselines—**Trans-MUSIC**, our reimplementation of TransMUSIC [15] and labelled Trans-MUSIC in every table and figure; **TCN-MUSIC**; and **DA-MUSIC**—are trained on independently generated one-bit datasets drawn from the same distribution and using the same sample counts and three training seeds, and decode $\hat{K}$ from their own source-number heads. All three read the raw sign sequence rather than a covariance, and differ in how they summarize it along time: Trans-MUSIC applies a Transformer encoder over the snapshot axis and mean-pools the result, TCN-MUSIC a temporal convolutional network followed by global average pooling, and DA-MUSIC a GRU whose final state is taken. In each case a linear layer maps the pooled feature to an $M \times M$ matrix that stands in for the covariance or its noise subspace—DA-MUSIC additionally eigendecomposes that matrix and selects the noise subspace with a learned gate—after which a MUSIC pseudospectrum is formed and an MLP regresses a fixed-length DOA vector that is truncated to $\hat{K}$. What the quantizer does to the statistics is therefore left for these networks to infer from data, whereas we invert it analytically in (4) and learn only a residual on top. In all three the regressor is trained with a permutation-minimizing RMSPE loss and the source-number classifier is fitted afterwards on frozen regressor features, so enumeration cannot shape the representation it reads; LCPSN

trains both heads jointly under (20). The classical block is the full cross product of three covariance front-ends and three enumeration rules. The front-ends are **FP-MUSIC**, which consumes the unquantized sample covariance and serves as a full-precision comparator under the same decoding metric; **one-bit MUSIC (naive)**, which applies MUSIC directly to the quantized sample covariance following the subspace-preservation strategy of One-bit MUSIC [11]; and **one-bit MUSIC (arcsine)**, which applies MUSIC to the arcsine-corrected covariance (5), i.e., the same statistic our network starts from. The enumeration rules are MDL and AIC [10] and an oracle- K rule that is given the true source number. We use the information-theoretic criteria rather than a hand-set eigenvalue threshold so that no baseline is disadvantaged by an arbitrary constant, and we include the oracle rows so that the cost of blind enumeration can be separated from the cost of quantization. MUSIC pseudospectra are evaluated on the same angular grid and decoded with the same peak-picking rule as the LCPSN spectrum. The covariance front end, source-enumeration rule, and spectrum estimator remain method-specific.

B. MAIN RESULTS

Table 1 summarizes the comparison. Four observations stand out.

Blind enumeration is the bottleneck of classical one-bit pipelines. With full-precision data, MDL attains 92.4% enumeration accuracy; under one-bit quantization, the same rule falls to 60.1% with the naive covariance and 25.0% with the arcsine-corrected covariance. AIC is worse for the naive one-bit covariance and no better than MDL for the arcsine-corrected covariance. The arcsine correction does not restore reliable blind enumeration in this implementation: both MDL and AIC yield approximately 25% accuracy on the balanced four-class test set. The oracle- K rows cannot establish an angular upper bound under minimum-error partial matching, as discussed below. These results suggest that the corrected covariance eigenstructure remains poorly matched to the standard enumeration criteria in the evaluated setting.

LCPSN restores reliable operation from one-bit data. When averaged over SNRs from 6 to 20 dB at $T = 200$, the proposed network reaches 97.36% enumeration accuracy, 1.07° MAE (1.53° RMSPE), and 93.80% SR@ 5° . At 10 dB and $T = 1000$, it reaches 0.76° MAE and 98.94% enumeration accuracy. The best blind one-bit classical pipeline has 3.86° MAE and 60.07% source-count accuracy, versus 1.07° and 97.36% for LCPSN. The strongest learned one-bit baseline has 2.46° MAE and 94.05% accuracy under the same protocol. Across the three reported training seeds, the LCPSN RMSPE spread is $\pm 0.007^\circ$, compared with the 1.56° difference to the nearest learned baseline in Table 1.

Figs. 5, 6, 7, and 8 compare learned baselines across SNR and snapshot sweeps.

LCPSN outperforms the evaluated full-precision MUSIC/MDL pipeline under partial matching. LCPSN beats FP-MUSIC/MDL on every column—enumeration by 4.95

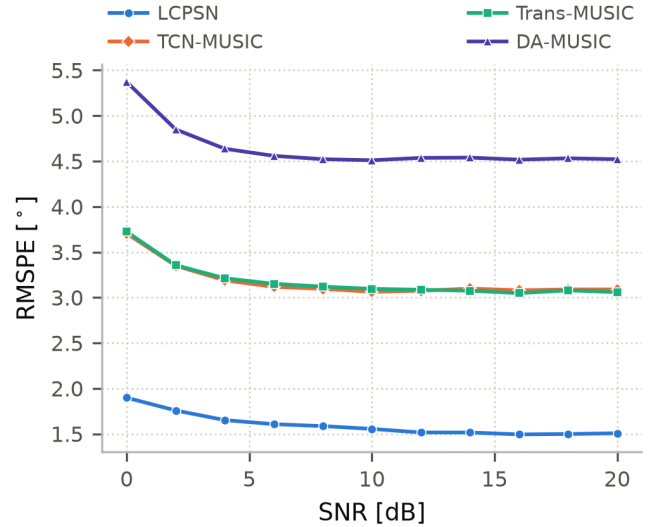


FIGURE 5. Matched RMSPE versus SNR at $T = 200$, LCPSN against the three learned one-bit baselines. Every curve is a mean over the same three training seeds. The sweep starts at 0 dB, the lower edge of the training SNR range.

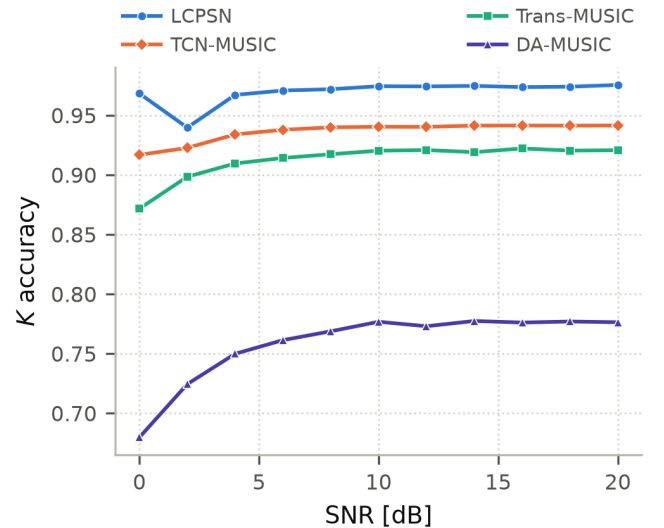


FIGURE 6. Source-number accuracy versus SNR at $T = 200$, same models and seeds as Fig. 5. The 2 dB point is the LCPSN enumeration dip discussed below; the baselines do not reproduce it.

points, MAE by $1.76\times$, SR@ 5° by 2.8 points—despite consuming one bit per component instead of full-resolution samples. This does not mean one-bit data carries more information than unquantized data; the comparison concerns the complete learned pipeline and the evaluated MUSIC/MDL implementation, and does not isolate the effect of the enumerator. With oracle K , FP-MUSIC has 2.99° partially matched MAE. This higher angular error than its MDL row is consistent with the fact that MDL undercounts are scored on fewer matched sources; it does not establish that oracle knowledge worsens the underlying estimator.

Performance is nearly flat within the training SNR

TABLE 1. Unknown- K DOA estimation on identical waveforms ($M=8$, $K \in \{2, \dots, 5\}$, 8° minimum separation, 25,600 scenes per operating point). The left block is the mean over the eight test SNR points satisfying $\text{SNR} \geq +6$ dB (6, 8, $\dots$, 20 dB) at $T=200$; the right block is evaluated at $T=1000$ and 10 dB. Learned rows are means over three training seeds; LCPSN across-seed standard deviations are reported in the text. FP-MUSIC consumes unquantized data for a full-precision comparison; oracle- K rows are given the true source number. Under partial matching, oracle- K angular errors need not improve on blind-rule errors. Best blind one-bit value in each column in bold.

Method	Input	Enum.	high SNR ($\geq +6$ dB), $T = 200$				$T = 1000$, 10 dB	
			K -acc (%)	MAE ($^\circ$)	RMSPE ($^\circ$)	SR@ 5° (%)	MAE ($^\circ$)	K -acc (%)
LCPSN (proposed)	1-bit	learned	97.36	1.07	1.53	93.80	0.76	98.94
TCN-MUSIC	1-bit	learned	94.05	2.46	3.09	85.10	2.07	97.03
Trans-MUSIC [15]	1-bit	learned	91.94	2.46	3.09	83.59	2.29	95.01
DA-MUSIC	1-bit	learned	77.33	3.72	4.53	67.25	4.34	70.38
One-bit MUSIC (naive)	1-bit	MDL	60.07	3.86	5.21	80.76	3.58	51.44
One-bit MUSIC (arcsine)	1-bit	MDL	25.00	6.10	8.12	67.00	4.70	25.00
One-bit MUSIC (naive)	1-bit	AIC	44.78	5.34	7.14	72.43	4.61	35.02
One-bit MUSIC (arcsine)	1-bit	AIC	25.00	6.10	8.12	67.00	4.70	25.00
FP-MUSIC	full prec.	MDL	92.41	1.89	2.65	90.96	1.53	92.81
FP-MUSIC	full prec.	AIC	87.80	2.27	3.17	88.78	1.95	88.24
One-bit MUSIC (naive)	1-bit	oracle K	100.00	6.32	8.52	66.88	4.95	100.00
One-bit MUSIC (arcsine)	1-bit	oracle K	100.00	6.42	8.62	67.22	4.88	100.00
FP-MUSIC	full prec.	oracle K	100.00	2.99	4.16	84.91	2.59	100.00

range. From 4 to 20 dB, LCPSN's enumeration accuracy remains within 96.7–97.5% and its MAE within 1.04° – 1.16° ; at 0 dB, the corresponding values are 96.8% and 1.34° . The 2 dB point is an exception: enumeration accuracy falls to 94.0% while MAE continues to decrease. The dip appears for all three LCPSN seeds and for the $\lambda_u = 0$ variant, but not for the three learned baselines evaluated on the same waveforms. This reduces the likelihood that the dip is specific to one test-set realization, but its cause within LCPSN remains unidentified. Within the plotted 0–20 dB training range, performance is comparatively insensitive to SNR. This conclusion does not extend below the training range: in the negative-SNR extrapolation regime, both enumeration and localization deteriorate sharply. At 10 dB, MAE decreases monotonically from 1.84° at $T = 50$ to 0.76° at $T = 1000$, and RMSPE decreases from 2.58° to 1.11° . Over the reported sweeps, changing T from 50 to 1000 at 10 dB changes MAE more than changing SNR from 0 to 20 dB at $T = 200$; the comparisons involve different ranges and do not isolate a general causal ranking.

The same four panels settle the comparison against the learned baselines. Beyond the high-SNR means of Table 1, they show that LCPSN's advantage holds at every plotted SNR from 0 to 20 dB and at every evaluated snapshot count rather than only in the mean, and that the ranking never crosses—in angular error (Figs. 5, 7) and in enumeration (Figs. 6, 8) alike. The snapshot panels also show where the improvement comes from: three of the four estimators improve monotonically with T , which is what the one-bit measurement model predicts, as more snapshots can reduce sampling error in estimated sign correlations under the independent-snapshot model. DA-MUSIC is the exception—its RMSPE rises from 4.51° at $T = 200$, the snapshot count it was trained at, to 5.29° at $T = 1000$, so its learned covariance surrogate does not generalize along T the way an

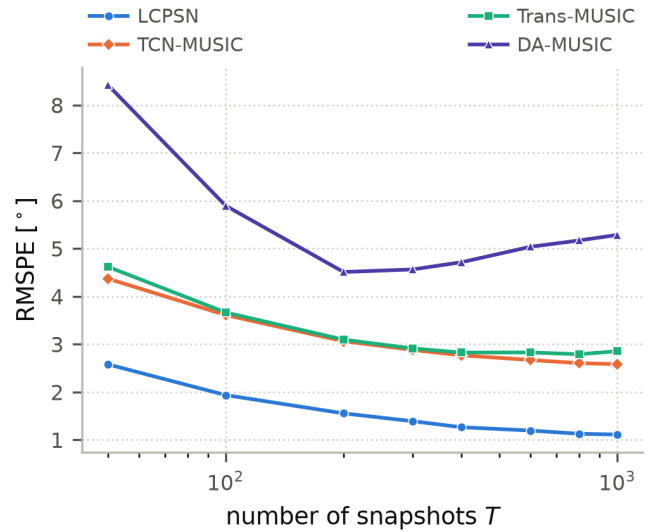


FIGURE 7. Matched RMSPE versus the number of snapshots at 10 dB. LCPSN improves monotonically with T , while DA-MUSIC degrades beyond $T = 200$.

explicitly recovered one does.

Figs. 9 and 10 compare the MDL-based MUSIC pipelines across SNR.

Figs. 9 and 10 plot the classical comparison. Two features are worth noting beyond the table. First, LCPSN sits below FP-MUSIC in RMSPE and above it in enumeration across the whole sweep, not only in the high-SNR mean. Second, the enumeration panel separates the two failure modes of the one-bit front-ends: the naive covariance degrades with SNR above 6 dB, while the arcsine-corrected front end remains near 25% accuracy from 2 dB upward.

The oracle- K rows of Table 1 complete the picture, and they read the opposite way to expectation: giving MUSIC the

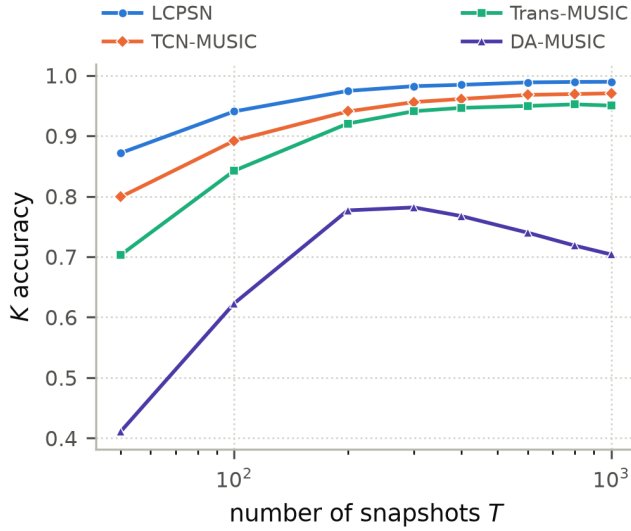


FIGURE 8. Source-number accuracy versus the number of snapshots at 10 dB, same models and seeds as Fig. 7.

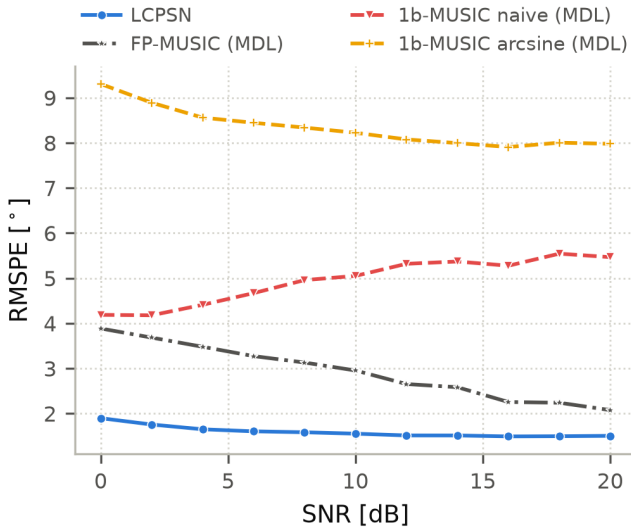


FIGURE 9. Matched RMSPE versus SNR at $T = 200$, LCPSN against classical MUSIC on three covariance front-ends. All classical rows use the blind MDL rule; FP-MUSIC consumes unquantized data and serves as a full-precision comparator under the stated partial-matching metric.

true source number makes its angular error worse for all three front-ends—one-bit naive $5.21^\circ \rightarrow 8.52^\circ$, one-bit arcsine $8.12^\circ \rightarrow 8.62^\circ$, full precision $2.65^\circ \rightarrow 4.16^\circ$ in RMSPE. This is the $\hat{K}$ -decoding property noted in Section V—a blind rule that under-counts is scored on fewer, easier angles—and it is why the oracle rows belong in the table as a diagnostic rather than as a stronger baseline.

Figs. 11, 12, and 13 show source-count confusion at 0, 10, and 20 dB.

Figs. 11, 12 and 13 show the enumeration confusion structure at 0, 10, and 20 dB and identify where the source-count errors occur. Essentially all error mass sits on the ± 1 off-

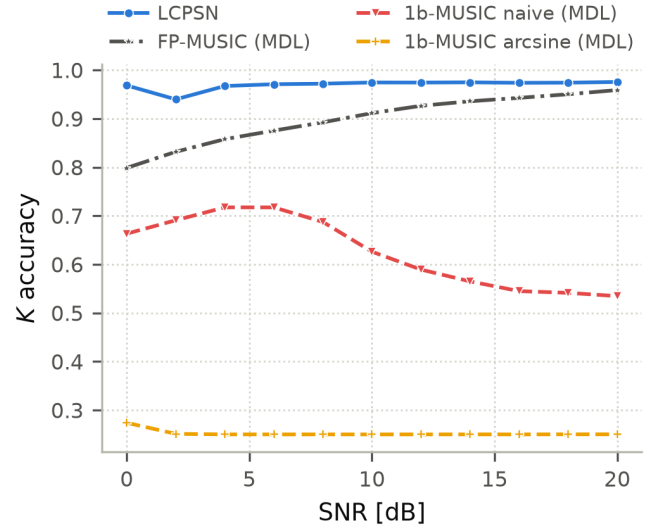


FIGURE 10. Source-number accuracy versus SNR at $T = 200$ for the same methods as Fig. 9. From 2 dB upward, the arcsine front-end remains at approximately 25% accuracy on the balanced four-class test set, indicating that the corrected covariance remains poorly matched to MDL-based source enumeration in this implementation.

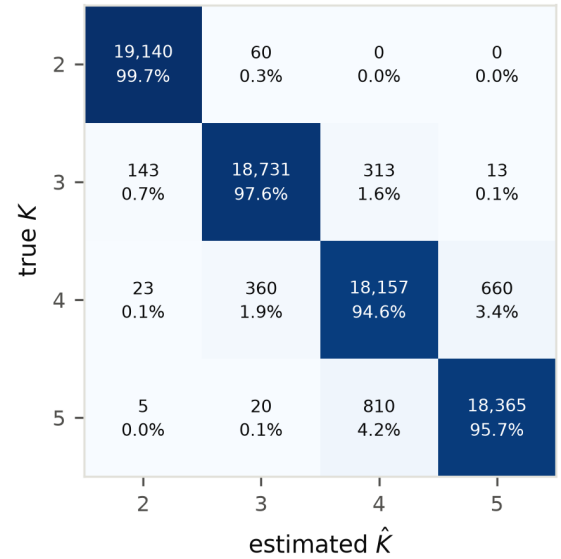


FIGURE 11. Source-number confusion of LCPSN at 0 dB, $T = 200$, pooled over the three training seeds (25,600 scenes per seed; cell text is the count and the row-normalized percentage). Error mass lies almost entirely on the ± 1 off-diagonal and concentrates on the dense $K = 4 \leftrightarrow 5$ pair.

diagonal: two-source scenes are recognized at 99.5–99.7%, and the residual confusion is concentrated between four and five sources, the densest pair an eight-element aperture has to separate. The two directions are of comparable weight—at 0 dB, 3.4% of four-source scenes are reported as five and 4.2% of five-source scenes as four—so the head is not systematically conservative; most errors occur in the denser source-count classes. Row accuracy falls with density from 99.7% at $K = 2$ to 94.6% at $K = 4$, then edges back up to 95.7% at $K = 5$ (all at 0 dB). The recovery at the

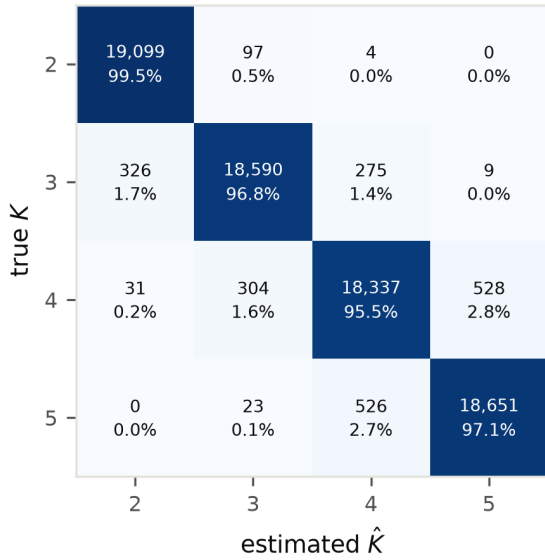


FIGURE 12. The same confusion structure at 10 dB. Overall accuracy rises from 96.87% to 97.24%, but the shape of the error remains similar.

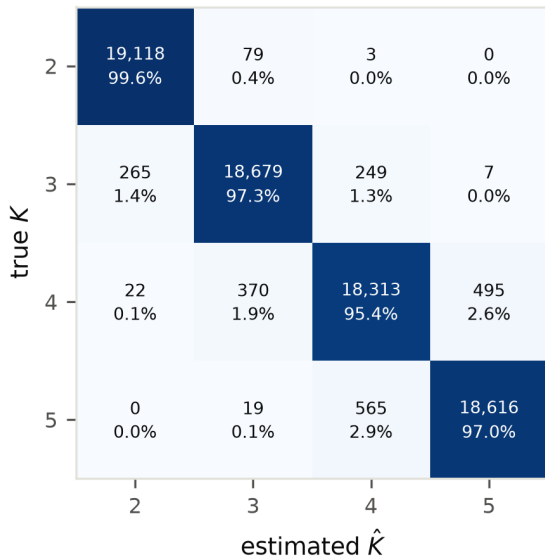


FIGURE 13. The same confusion structure at 20 dB (97.30%). Across the 0–20 dB range, the off-diagonal mass remains concentrated on the ± 1 band, which is consistent with source-density effects but does not isolate array resolution from other causes.

top of the range is an artifact of the bounded label set: with $\mathcal{K} = \{2, \dots, 5\}$ a five-source scene can only be miscounted downward and a two-source scene only upward, so the two boundary classes each have one fewer way to be wrong. The interior classes carry the real difficulty, and there it grows with density, consistent with closely spaced sources merging under the resolution limit of an eight-element array observed through one-bit data.

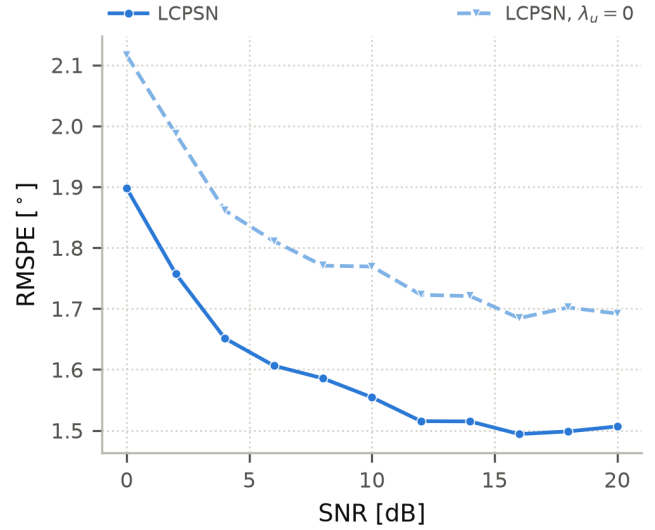


FIGURE 14. Ablation of $\mathcal{L}_{\text{ucov}}$: matched RMSPE versus SNR at $T = 200$, three seeds, everything but λ_u held fixed. The weighted model is better at every point in the sweep.

TABLE 2. Ablation of the $\mathbf{U}_0$ -weighted covariance consistency (identical training protocol and seeds; mean over three seeds, high SNR $\geq +6$ dB at $T=200$). Here, $\lambda_u = 0$ removes the uncertainty-weighted consistency term, whereas $\lambda_u = 10^{-3}$ is the default configuration. Bold marks the better configuration; the K -accuracy column is left unmarked because its 0.01-point difference is below the seed spread.

λ_u	K -acc (%)	MAE ($^\circ$)	RMSPE ($^\circ$)	SR@5 $^\circ$ (%)
0	97.37	1.21	1.73	92.72
10^{-3}	97.36	1.07	1.53	93.80

C. ABLATION: UNCERTAINTY-WEIGHTED CONSISTENCY

Table 2 isolates the contribution of $\mathcal{L}_{\text{ucov}}$ by setting $\lambda_u = 0$ with every other setting and all three seeds unchanged. The effect separates along the two outputs.

The measured enumeration change is within seed variability: 97.36% versus 97.37%, a difference of 0.01 points against a seed spread of 0.04. The angular outputs, in contrast, improve throughout: MAE $1.21^\circ \rightarrow 1.07^\circ$ (−11.4%), RMSPE $1.73^\circ \rightarrow 1.53^\circ$, and SR@5 $^\circ$ 92.72% $\rightarrow$ 93.80%. Figs. 14 and 15 show that the weighted model is better at all 11 SNR points from 0 dB up and at all 8 snapshot counts, in every case by more than the pooled seed noise, so the high-SNR mean is not being carried by one favourable operating point.

$\mathcal{L}_{\text{ucov}}$ anchors $\mathbf{R}_{\text{ref}}$ to $\mathbf{R}_0$ entry-by-entry with a stiffness set by $\mathbf{U}_0$. Because both the spectrum and the eigen-features consumed by the K -head are computed from $\mathbf{R}_{\text{ref}}$, this term can in principle influence both localization and enumeration. Empirically, however, the measured change in enumeration accuracy remains within across-seed variability, whereas the angular improvement is consistent across all evaluated operating points. We therefore interpret the dominant measured effect as improved localization.

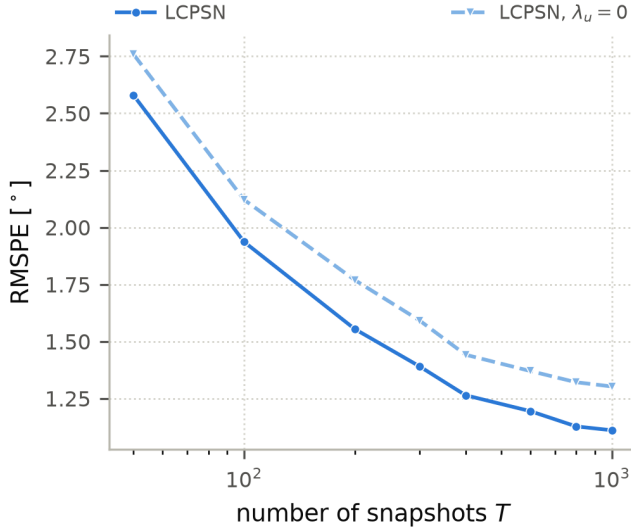


FIGURE 15. The same ablation swept over snapshots at 10 dB. The gap persists across a twentyfold change in T , so the weighting is not compensating for one particular sample size.

VI. CONCLUSION

We presented LCPSN, a learnable covariance probabilistic spectrum network for one-bit DOA estimation that operates without knowledge of the source number. The method combines arcsine covariance recovery, Transformer-based Hermitian residual refinement, probabilistic spectrum heads driven by a Bartlett prior spectrum, and an eigen-feature source-number head, and—as its distinguishing ingredient—uses a closed-form, first-order approximation to the finite-snapshot uncertainty of the one-bit arcsine-corrected correlation estimate as an entry-wise weight in a covariance-consistency objective. On blind one-bit benchmarks with $K \in \{2, \dots, 5\}$ unknown sources, LCPSN attains 97.4% enumeration accuracy and 1.07° MAE, improving to 0.76° at $T = 1000$, exceeding both classical one-bit MUSIC and learned one-bit baselines by wide margins and surpassing even full-precision MUSIC under the same blind protocol.

The comparison in Table 1 deliberately couples two difficulties—quantization and blind enumeration—because they interact: quantization distorts exactly the eigen-structure that classical enumeration consumes. LCPSN combines a K -head that learns refined covariance eigen-features, a false-peak penalty (14) that penalizes spurious spectrum responses, and a U_0 -weighted covariance-consistency term. The ablation over three seeds evaluates the last component: removing $\mathcal{L}_{ucov}$ increases the observed MAE by 0.14° , while the source-count difference is within across-seed variability. The experiment isolates the contribution of uncertainty weighting to the measured angular improvement, but does not separately ablate the K -head or false-peak penalty.

Future work includes coherent and calibrated-hardware scenarios, gridless decoding in the spirit of [15], dithered (time-varying threshold) acquisition [25], [16] for amplitude identifiability, and extension to two-dimensional (azimuth–

elevation) arrays.

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APPENDIX A. QUANTIZER SCALING CONVENTION

Some works, e.g., [15], include a factor of $1/\sqrt{2}$ in the quantizer (2) so that $|y_m(t)| = 1$. Our convention omits it without loss of generality: for any $c > 0$, replacing $\mathbf{y} \rightarrow c\mathbf{y}$ scales $\mathbf{R}_z \rightarrow c^2\mathbf{R}_z$ and $\mathbf{D} \rightarrow c^2\mathbf{D}$ in (5), leaving $\mathbf{C}_z$ and hence $\mathbf{R}_0$ unchanged.

APPENDIX B. STEERING GRID OF THE BARTLETT PRIOR SPECTRUM

The steering matrix used to evaluate the Bartlett prior spectrum (9) is sampled on the half-open variant $-90^\circ + g \cdot (180^\circ/G)$ instead of the closed grid of Section IV. The prior is processed by learned spectrum heads; although the target and decoder use the same closed grid, the effect of the prior-grid mismatch on accuracy has not been separately measured. Aligning these grids is a straightforward implementation improvement.

APPENDIX C. VARIANCE SPLIT OF THE REAL-AUGMENTED COVARIANCE ENTRIES

The factor $\frac{1}{4}$ in $\mathbf{V}_{\Re}$ and $\mathbf{V}_{\Im}$ follows from the real-augmented assembly of the complex entry, $\Re R_{ij} = \frac{1}{2}(\hat{C}_{rr} + \hat{C}_{ii})_{ij}$ and $\Im R_{ij} = \frac{1}{2}(\hat{C}_{ir} - \hat{C}_{ri})_{ij}$: if the two contributing block estimates are treated as uncorrelated, the variance of a half-sum (or half-difference) is one quarter of the sum of their variances. Sign products sharing the same snapshots can be

correlated, so these formulas define approximate diagonal weights rather than exact variances of the complex entries.

HYUK KWON is currently pursuing the B.S. degree in computer and information engineering at Kwangwoon University, Seoul, Republic of Korea. His research interests include machine learning, computer vision, and artificial intelligence.

HANASOL JEON is currently pursuing the B.S. degree in computer and information engineering at Kwangwoon University, Seoul, Republic of Korea. His research interests include cryptography, Internet of Things (IoT) security, and cyber-physical system (CPS) security.

YOUNGIN CHO is currently pursuing the B.S. degree in computer and information engineering at Kwangwoon University, Seoul, Republic of Korea. His research interests include artificial intelligence, machine learning, and backend development for software services.

GYUNGMOCK KIM is currently pursuing the B.S. degree in computer and information engineering at Kwangwoon University, Seoul, Republic of Korea. His research interests include artificial intelligence security, quantum cryptography, and metaverse privacy.

HO YOUNG HWANG (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering and computer science from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, in 2000, 2002, and 2008, respectively.

From 2002 to 2003, he was a Guest Researcher with the Advanced Network Technologies Division, National Institute of Standards and Technology (NIST), Gaithersburg, MD, USA. From 2008 to 2010, he was a Postdoctoral Fellow with the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON, Canada. He joined the faculty of Kwangwoon University, Seoul, Republic of Korea, in 2010, where he is currently a Professor with the School of Computer and Information Engineering. His research interests include wireless networks, embedded systems, sensor networks, machine learning, communication systems, protocols, and services.

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